PORTFOLIO PROJECT · PROCESS CONTROL

Blast-furnace top-pressure control with a top-gas recovery turbine

A turbine recovers ≈ 6.9 MW from blast-furnace gas, yet must be invisible to the furnace. I designed and simulated the control system that keeps top pressure within **17 mbar** when the turbine trips at full load.

Control narrative | P&ID · I/O list · C&E | Simulink physics model | Split-range PI · feedforward | IEC 61511 note

17 mbar
top-pressure peak at a full-load trip, with feedforward

214 mbar
same trip, feedback only

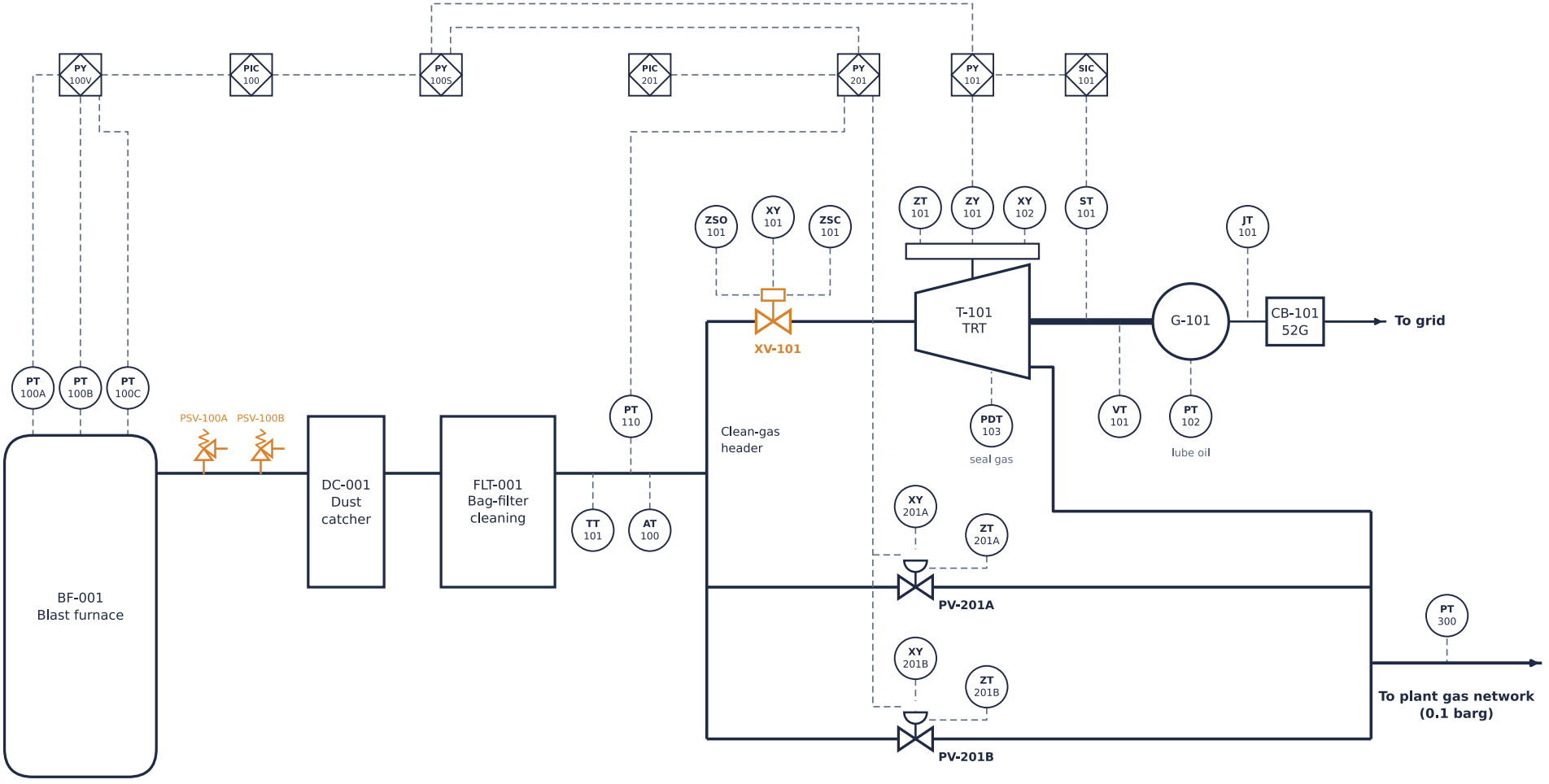
6.9 MW
recovered at design point

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Code and documents: **github.com/zano0/trt-top-pressure-control**

THE PROBLEM

The turbine earns money only if the furnace never notices it

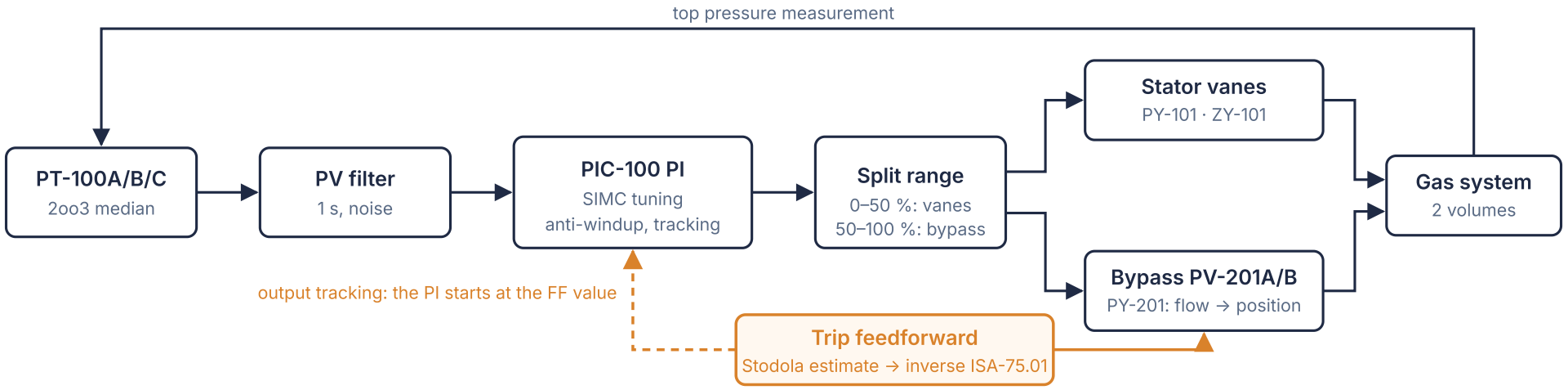
Top gas (280 000 Nm³/h, 1.8 barg) is expanded in the turbine T-101 to recover **6.9 MW**; bypass valves PV-201A/B in parallel take the gas when it cannot. Top pressure belongs to the furnace and must stay at set point. The hard case is a **trip**: XV-101 closes in 0.7 s and 104 kg/s must move to the bypass, or pressure climbs ≈ 30 mbar/s toward the 2.2 barg bleeders. Target **≤ 90 mbar**.



Simplified P&ID (ISA-5.1), vector: zoom in the PDF. Generated from the single tag list that also produces the I/O list and the cause & effect matrix; legend and notes in `docs/pid/`.

CONTROL STRATEGY

One pressure controller, two actuators, and a feedforward for the event feedback cannot catch



Split range

The vanes act first, so all the gas goes through the turbine. The bypass opens only when the vanes saturate. Bypass requests are in **flow**, so the loop gain is the same on both sides.

Anti-windup & bumpless

Back-calculation on the position actually reached; external reset during the high-high override. Every mode change re-initialises the PI.

Trip feedforward

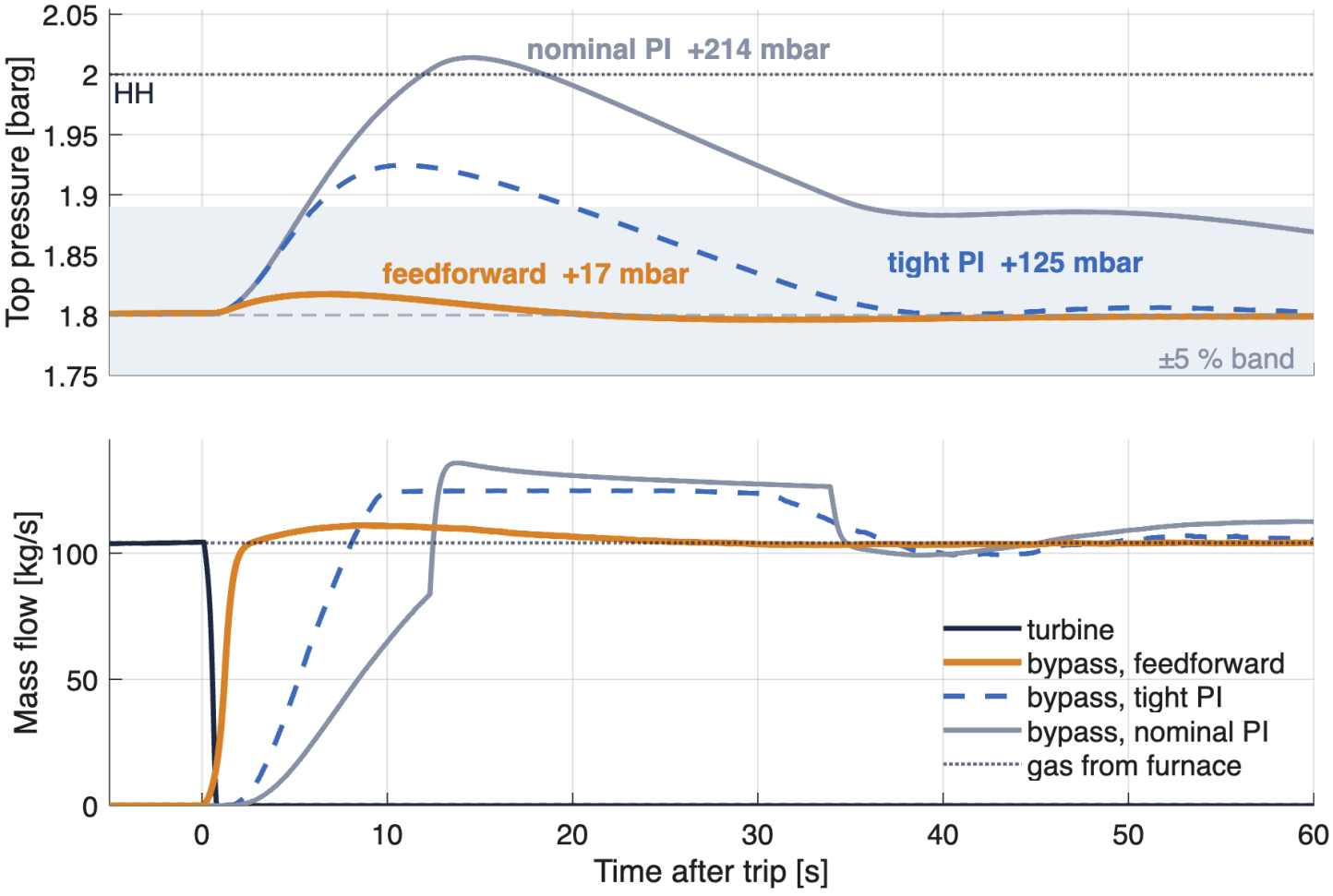
On the trip signal the bypass goes straight to the position that passes the turbine flow just before the trip. The bypass is choked, so a rising pressure adds relief by itself.

Tuned from the model

SIMC rules on the linearised two-volume plant (84 s and 2.6 s time constants):
Kp = 164 %/bar, Ti = 38 s.
Gain scheduling for bypass-only operation.

KEY RESULT

At a full-load trip the feedforward holds top pressure within 17 mbar; feedback alone cannot get close



17 mbar

peak with feedforward + output tracking (target 90)

214 mbar

nominal PI, feedback only: reaches the high-high level

125 mbar

tightest recommended PI (SIMC $\tau c = \theta$) with a light filter

To reach 90 mbar, feedback alone needs a phase margin of only 20°, and it is still 5× worse. The gas disappears in 0.7 s; the loop sees it only after 2–3.6 s of effective dead time. Robust to actuator speed (7–27 mbar for ESD 0.5–1 s, bypass 1–2 s) and to a ±20 % feedforward error (≈ 50 mbar).

SCOPE AND HONESTY

What exists today, what the model leaves out, what a real project adds

Delivered

- Control narrative in EPC style: modes, loop assignment per mode, sequences, trips
- P&ID, I/O list (31 points), alarm list and cause & effect, all generated from one tag list
- Design basis with a sourced, consistent design point
- Simulink model generated by script; codegen-ready PLC logic
- Scenarios with KPIs: steps **28 mbar**, slip **56 mbar**, loading **3.5 mbar**, trip **17 mbar**
- Functional-safety note (IEC 61511, hypothetical SIL)

Limits of the model

- Illustrative parameters, no plant data
- Constant turbine efficiency, linear vane characteristic, no stage-by-stage model
- Isothermal gas, stiff downstream network
- Simplified loading sequence; synchronisation and the PLC implementation come next
- No hardware: tests are simulated, not a factory acceptance test

In a real project I would

- Fit the turbine and valve characteristics to vendor data and step tests
- Agree the interface and the scope split with the blast-furnace DCS
- Move overspeed to an API 670 system and SIL-rated functions to a safety PLC
- Validate the feedforward in commissioning with partial-load trips

Next in this repository

- IEC 61131-3 Structured Text on OpenPLC, Software-in-the-Loop over Modbus, HMI
- MPC on a 24 h disturbance profile

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